Problem Solving - Binary Search

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Problem 1: Ropes

n ropes :

$a = [\ a_1\ ,\ a_2\ \dots\dots\ a_n\ ]$

$a_i \rightarrow$ length of ith rope.

(K pieces of rope) → equal length ,, .

max length x

T T T T T

1 2 3 4 5

l r

low, high

$mid = (low + high)/2$

$a_1 = 12$

$x = 5$

2

2.4

0.5

mid → k pieces

$a = [a_1, a_2, a_3, \ldots, a_n]$

$\lfloor a_1/x \rfloor \; \lfloor a_2/x \rfloor \; \lfloor a_3/x \rfloor \; \ldots \; \lfloor a_n/x \rfloor$

0.003

low ____ mid ____ high

min → low

low = 1

$n = k$

$\frac{1}{100} \to 0.01$

$a = [1, 1, 1] \to k = 300$

count = 0

for (1 → n) :
count += a[i]/x

→ if (count >= k) return true ;
→ return false;

$\boxed{> K}$

$\boxed{\text{count} == K}$

count > K

low = 0 → mid = 0

mid = (low + high)/2 = $\boxed{0}$

0

1 . /2

(0 + 1)/2 = 0.5

(0 + 0.5)/2 → 0.25

0.125

× 0

no. of iterations,, $\rightarrow$ 10^{-6} $\rightarrow$ 6 decimal places,,

$\log_2((high - low) \times 10^{6})$ $\rightarrow$ 6,,

low = 0, high = 10^2

K = 1 $\quad a = [10^2 \ldots\ldots 10^2]$ $\quad \log_2(10^2 \times 10^6)$

$\boxed{x = 10^2}$

$\log_2(10^{13})$

$13 \times \log_2(10) \rightarrow (3.322 \times 13)$

43.186 $\rightarrow$ 44 $\rightarrow$ 50

$0(\_\ \_\ \_\ \_\ \_\ \_)$

$10^{-1}\ 10^{-2}\ 10^{-3}\ 10^{-4}\ 10^{-5}\ 10^{-6}$

$\boxed{9 \times 10^{-2}} < 10^{-6}$

4 2

12.125383 9 5 7

125384 10^{-6}

cout <<

set

Setprecision(7)

39 → 40

Printf("%.7lf", ans) 10^{-7}

Problem 1
Solution Code

```cpp
bool checker(vector<int> &ropes, long double val, int k) {
    long long count = 0;

    for(auto it : ropes) {
        count += (it / val);
    }

    return count >= k;
}

long double binSearch(vector<int> &ropes, int k) {

    long double low = 0, high = 1e7;

    for(int it = 0; it < 50; it++) {
        long double mid = (low + high) / 2;

        if(checker(ropes, mid, k)) {
            low = mid;
        }
        else {
            high = mid;
        }
    }

    return low;
}
```

Problem 2: Jumping Through Segments

Main Idea:

- Perform binary search on the value **k**.
- Calculate the minimum and maximum coordinate you can reach on the ith iteration while also maintaining the condition that you are still inside the coordinates of the ith segment for a given **k**.
- If no such range is possible, then the player cannot complete the level with given **k**.

```cpp
bool checker(vector<array<long long, 2>> &segments, int n, long long val) {
    long long posMin = 0, posMax = 0;

    for(int i = 0; i < n; i++) {
        posMin = max(posMin - val, segments[i][0]);
        posMax = min(posMax + val, segments[i][1]);

        if(posMin > posMax) return false;
    }

    return true;
}

long long binSearch(vector<array<long long, 2>> &segments, int n) {
    long long low = 0, high = 1e9;

    long long ans = 0;

    while(low <= high) {
        long long mid = (low + high) / 2;

        if(checker(segments, n, mid)) {
            ans = mid;
            high = mid - 1;
        }
        else {
            low = mid + 1;
        }
    }

    return ans;
}
```

Problem 2 Solution Code

vector<pair>

k

n iterations

l_i x r_i

2nd

1st

ith

$k \le ((l_i \le x \le r_i))$

$-k$

$[-k, k]$

1st 2nd 3rd

$1 \to n$

x<?

low

T

T

K

T T T T T

mid

larger

F F F

K

high

→ True

high = mid

→ False

low = mid

l_1 r_1 l_2 r_2

K = 5

K = 6

$x = 0 \to l_1 \qquad r_1$

$[-k, k]$

1^{st}

k

$minPos_N = \max(l_1, -k)$

$maxPos = \min(r_1, k)$

$(i-1)^{th}$

$minPos = -1e18 \qquad max = 1e18$

$$\boxed{\begin{aligned} minPos &= \max(minPos - k, l_i) \\ maxPos &= \min(maxPos + k, r_i) \end{aligned}}$$

After $(i-1)$ iterations → $\boxed{minPos > maxPos}$ → false

$[minPos, maxPos]$

After ith iteration:

→ $[minPos - K, maxPos + K]$

$[l_i, r_i]$

ith $==$ $<=$ → True

intersections

$\{ \_ \}$ $\boxed{\text{ith segment}}$

minPos → _ $>$

maxPos → _ $=$

$(l > r)$

ith

Problem 3: Get Together

Main Idea:

- Perform binary search on the time required.
- For a given time, calculate the minimum and maximum coordinate that can be reached for every person.
- If there is no coordinate in the intersection of all these ranges, then the people cannot gather at a single point in given time.

```cpp
bool checker(vector<array<long long,2 >> &persons, long double time) {
    long double minX = -1e18, maxX = 1e18;

    for(auto &[initialPos, velocity] : persons) {
        minX = max(minX, initialPos - (velocity * time));
        maxX = min(maxX, initialPos + (velocity * time));

        if(minX > maxX) return false;
    }

    return true;
}

long double binSearch(vector<array<long long, 2>> &persons) {
    long double low = 0, high = 1e9;

    for(int it = 0; it < 60; it++) {
        long double mid = (low + high) / 2;

        if(checker(persons, mid)) {
            high = mid;
        }
        else {
            low = mid;
        }
    }

    return high;
}
```

Problem 3 Solution Code

n people

x_i v_i

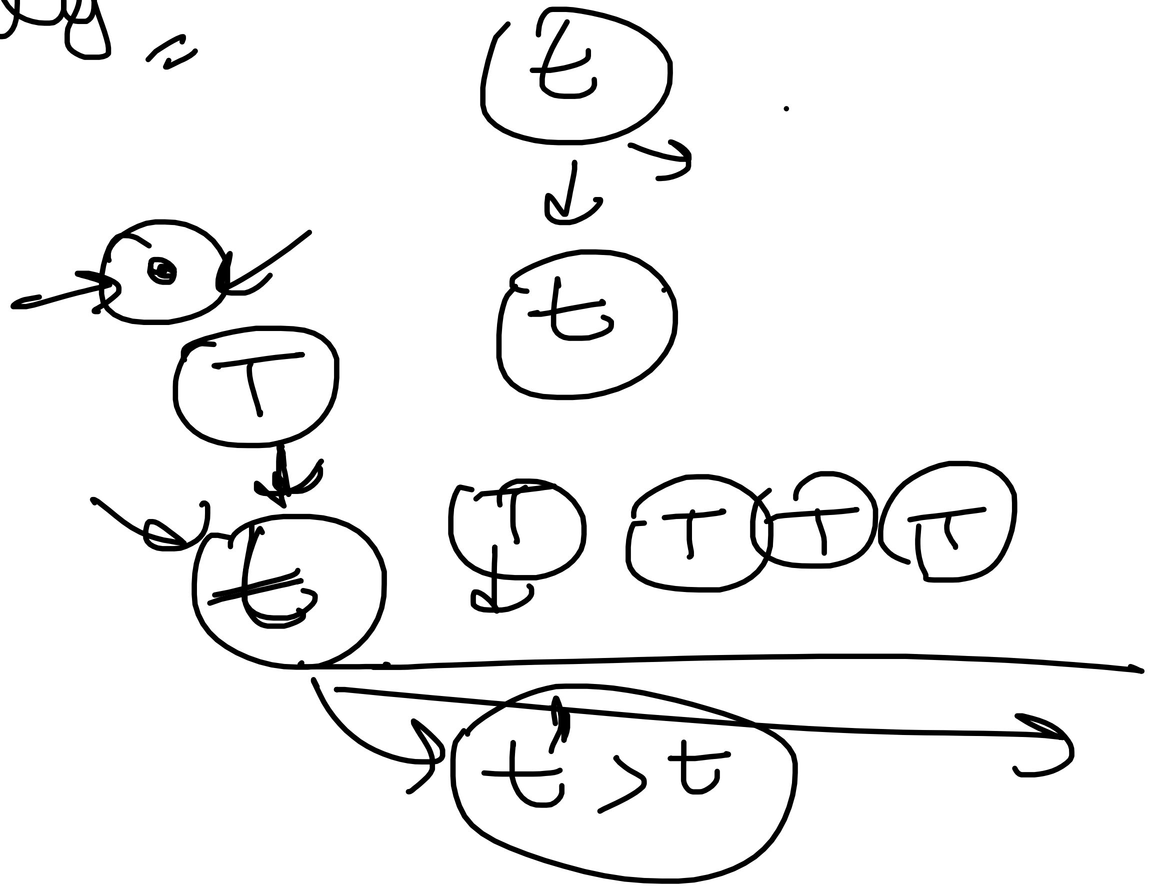
→ velocity

initial pos

$t = 4s$

5

$t = 3s$



t

$$\left|\left[x_i - v_i \times t,\ x_i + v_i \times t\right]\right|$$

$$l \qquad\qquad r$$

$$\left[(l_1,\ r_1)\ (l_2,\ r_2),\ (l_3,\ r_3)\ (l_n, r_n)\right]$$

Intersection

$[1, 5]$, $[2, 5]$, $[-3, 0]$

$t = 2.5$

$-3 \quad 0 \quad 1 \quad 2 \quad 5$

pos

$minX \le maxX$

$low = 0$

$high = 1e9$

$n = 10$

$\log_2((high - low) \times 10^6)$

O

$\{[-10^9, 1], [10^9, 1]\} \rightarrow \log_2(10^{15})$

$\boxed{10^9}$ $\boxed{10^9}$

$maxX = \min($

$minX = \max($

$(minX, maxX)$

$minX \quad l \quad maxX \quad r$

$(minX, maxX) \leftarrow (l, r)$

i

t

$(\max(l, minX),$

$\min(r, maxX))$